

2024-09-05-SimplexEfficiency

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Slide 1

Content

Efficiency of Simplex Method

- Worst case vs Average Case
- (Dantzig) $m < 50$, $m+n < 200$

$$\#(\text{iterations}) \leq \frac{3m}{2} \quad (\text{usually}) \quad \text{rarely} \approx 3m$$

Description

This slide discusses the efficiency of the simplex method in linear programming. It contrasts the worst-case and average-case performance. The worst-case scenario, attributed to Dantzig, is characterized by problem dimensions where $m < 50$ and $m + n < 200$, where m is the number of constraints and n is the number of variables. The average number of iterations required is typically bounded above by $\frac{3m}{2}$, although in rare cases, it might approach $3m$. Two tables (Table 4.1 and Table 4.2) present empirical data on the average number of iterations needed for different problem sizes (m and n) using two different pivot rules: the largest-coefficient rule and the largest-increase rule. The context of the course (Linear Programming) is crucial for understanding the practical implications of these efficiency considerations when choosing a solution method for linear programming problems.

Figures

Efficiency of Simplex Method
• Worst case vs Average Case

(a) Efficiency of Simplex Method title and bullet points.

Worst Case
Dantzig
#(iterations)

[c]

n	Average Number of Iterations
10	10.0
20	20.0
30	30.0
40	40.0
50	50.0
60	60.0
70	70.0
80	80.0
90	90.0
100	100.0

(b) Table 4.1

Average Case
 $m < 50$, $m+n < 200$
 $\frac{3m}{2}$, rarely $3m$

n	Average Number of Iterations
10	10.0
20	20.0
30	30.0
40	40.0
50	50.0
60	60.0
70	70.0
80	80.0
90	90.0
100	100.0

(c) Table 4.2

Slide 2

Content

Worst Case Example: Klee-Minty (1970) [v]

The example is quite simple to state:

$$\begin{aligned} &\text{maximize } \sum_{j=1}^n 2^{n-j} x_j \\ (4.1) \quad &\text{subject to } \sum_{j=1}^{i-1} 2^{i-1-j} x_j + x_i \leq 100^{i-1} \quad i = 1, 2, \dots, n \\ &x_j \geq 0 \quad j = 1, 2, \dots, n. \end{aligned}$$

It is instructive to look closely at the constraints. The first three constraints are

$$x_1 \leq 1$$

$$4x_1 + x_2 \leq 100$$

$$8x_1 + 4x_2 + x_3 \leq 10,000.$$

Description

This slide presents a worst-case example for the simplex method, attributed to Klee and Minty (1970). The example is a linear program in n variables, aiming to maximize an objective function that is a weighted sum of the variables. The weights are powers of 2. The constraints are structured such that the feasible region is a deformed hypercube. The slide highlights the constraints, showing the first three explicitly to illustrate the pattern. The problem is designed to demonstrate that the simplex method can take an exponential number of iterations in the worst case, contrasting with its typically good average-case performance. The notation (4.1) suggests this is equation 4.1 in the textbook or lecture notes. The context of the course (Linear Programming) is crucial for understanding the implications of this worst-case example on the efficiency and limitations of the simplex method.

Figures

(a) Worst Case Example title and author.

(b) Linear Programming problem statement.

(c) First three constraints.

Slide 3

Content

Worst Case Example: Klee-Minty (1970)

[v]

$$\text{maximize } \sum_{j=1}^n 2^{n-j} x_j - \frac{1}{2} \sum_{j=1}^n 2^{n-j} \beta_j$$

$$(4.2) \text{ subject to } 2 \sum_{j=1}^{i-1} 2^{i-1-j} x_j + x_i \leq \sum_{j=1}^{i-1} 2^{i-1-j} \beta_j + \beta_i \quad i = 1, 2, \dots, n$$

$$x_j \geq 0 \quad j = 1, 2, \dots, n$$

$$1 \ll \beta_1 \ll \beta_2 \ll \dots \ll \beta_n$$

Largest Coefficient Rule: require $2^n - 1$ steps

Smallest Coefficient Rule: just 1 steps

Description

This slide presents a worst-case example for the simplex method, the Klee-Minty problem (1970). The linear program aims to maximize a weighted sum of variables x_j , with weights being powers of 2, and subtracting another weighted sum of parameters β_j . The constraints are structured to create a highly skewed feasible region. The notation (4.2) suggests this is equation 4.2 in the textbook or lecture notes. The slide then highlights the impact of different pivot rules on the number of iterations required to reach the optimal solution. The "Largest Coefficient Rule" requires an exponential number of steps ($2^n - 1$), while the "Smallest Coefficient Rule" requires only one step. The notation $1 \ll \beta_1 \ll \beta_2 \ll \dots \ll \beta_n$ indicates that the β_i values are strictly increasing and significantly different in magnitude. This example demonstrates that the simplex method's efficiency can be highly dependent on the pivot rule used and that in the worst case, it can exhibit exponential time complexity. The context of the course (Linear Programming) is crucial for understanding the implications of this worst-case example on the efficiency and limitations of the simplex method.

Figures

$$(4.2) \quad \text{subject to } 2 \sum_{j=1}^{i-1} 2^{i-1-j} x_j$$

(a) (Inequality signs)

$$(4.2) \quad \text{subject to } 2 \sum_{j=1}^{i-1} 2^{i-1-j} x_j$$

(b) (Largest and Smallest Coefficient Rules)